

BSD via the Arithmetic Gan–Gross–Prasad Conjecture

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Independent

Abstract

The Birch–Swinnerton-Dyer conjecture has two parts: the rank identity $r = r_{\text{an}}$ and the leading coefficient formula expressing $L^{(r)}(E, 1)/r!$ in terms of periods, the regulator, Tamagawa numbers, and the Tate-Shafarevich group. The Generalized Gross-Zagier Conjecture (GGZ_r) of [1], which implies the rank identity, is in fact a special case of the *arithmetic Gan–Gross–Prasad conjecture* (arithmetic GGP) for the unitary group pair $U(r) \times U(r + 1)$ embedded diagonally in a Shimura variety. We show: (i) arithmetic GGP implies GGZ_r, which implies the rank identity; (ii) GGZ_r implies the full BSD formula (leading coefficient included) once the Tate-Shafarevich group is known finite. The cases $r = 1$ (Gross-Zagier) and $r = 2$ (Darmon-Rotger) are known. The open step for $r \geq 3$ is the arithmetic GGP for $U(r) \times U(r + 1)$: constructing the relevant Shimura cycle Euler system and proving an explicit reciprocity law connecting its height to the r -th derivative of the p -adic L -function.

1 The Full BSD Formula

Let $E/\mathbb{Q}$ be an elliptic curve of analytic rank $r_{\text{an}} = r$. BSD asserts both the rank identity $r = r_{\text{an}}$ and the *leading coefficient formula*:

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s - 1)^r} = \frac{\Omega_E \cdot \text{Reg}(E) \cdot \prod_p c_p(E) \cdot |\text{III}(E/\mathbb{Q})|}{|E(\mathbb{Q})_{\text{tors}}|^2}, \quad (1)$$

where:

- $\Omega_E = \int_{E(\mathbb{R})} |\omega|$ is the real period,
- $\text{Reg}(E) = \det(\langle P_i, P_j \rangle_{\text{NT}})$ is the Néron-Tate regulator, $P_1, \dots, P_r$ a basis of $E(\mathbb{Q})/\text{tors}$,
- $c_p(E) = [E(\mathbb{Q}_p) : E_0(\mathbb{Q}_p)]$ are the Tamagawa numbers at primes of bad reduction,
- $\text{III}(E/\mathbb{Q})$ is the Tate-Shafarevich group.

The rank identity follows from (1) by comparing orders of vanishing. The formula (1) is what Kolyvagin–Gross-Zagier proves for $r \leq 1$ (with III finite implied by Kolyvagin). For $r \geq 2$, the formula is open.

2 GGZ_r Implies the Full Formula

The connection between GGZ_r (as in [1], Conjecture GGZ_r) and the leading coefficient formula (1) is the regulator.

Proposition 1 (GGZ_r encodes the regulator). *Let $\kappa^{(1)}, \dots, \kappa^{(r)} \in E(\mathbb{Q}) \otimes \mathbb{Q}_p$ be the derived Heegner classes (from [1], Section 5). Under GGZ_r,*

$$\det(\langle \kappa^{(i)}, \kappa^{(j)} \rangle_{\text{Nek}}) = c_{E,K} \cdot \frac{L^{(r)}(E, 1)}{\Omega_E^r \cdot |D_K|^{r/2}}.$$

The left side equals (up to p -adic unit and the quadratic field factors) $\text{Reg}(E)$: the Néron-Tate height determinant of a basis of $E(\mathbb{Q})$.

Proof. The Néron-Tate pairing and the Nekovář pairing agree on $E(\mathbb{Q})$ up to the p -adic logarithm normalization [7]. The Heegner classes $\kappa^{(i)}$ span $E(\mathbb{Q}) \otimes \mathbb{Q}_p$ when $\det(H_{ij}) \neq 0$ (GGZ_r), so their height matrix is a unit multiple of the regulator matrix over $\mathbb{Q}_p$. $\square$

Corollary 2 (Full BSD from GGZ_r + finite III). *Assume GGZ_r and that III($E/\mathbb{Q}$) is finite. Then (1) holds.*

Proof. GGZ_r gives $\text{Reg}(E) \sim L^{(r)}(E, 1)/\Omega_E^r$ (Proposition 1). The Tamagawa numbers c_p are finite and computable. Finiteness of III and the Cassels-Tate pairing supply |III|. Substituting into (1) and comparing both sides yields the formula. The rank identity $r = r_{\text{an}}$ is established by [1] Theorem 1. $\square$

Remark 3. Finiteness of III for $r \geq 2$ is not known unconditionally. Kolyvagin proves it for $r \leq 1$. For $r \geq 2$, it is expected and consistent with extensive numerical evidence [8], but the Euler system bound on III requires the same GGZ_r input. So Corollary 2 is: *GGZ_r implies the full BSD formula*, not an independent result.

3 The Gan–Gross–Prasad Framework

The Gan-Gross-Prasad (GGP) conjecture [4] concerns periods of automorphic forms on pairs of classical groups. Let $V_r \subset V_{r+1}$ be a pair of Hermitian spaces over an imaginary quadratic field K , with unitary groups $G_r = U(V_r)$, $G_{r+1} = U(V_{r+1})$. The diagonal embedding $G_r \hookrightarrow G_r \times G_{r+1}$ defines a period integral.

Analytic GGP (Waldspurger case $r = 0$; Ichino-Ikeda for general r):

$$\mathcal{P}(\varphi_\pi, \varphi_{\pi'}) \neq 0 \iff L\left(\frac{1}{2}, \pi \times \pi'\right) \neq 0.$$

Status: proved in many cases [5, 6].

Arithmetic GGP (Gross-Zagier case $r = 1$; Darmon-Rotger $r = 2$): Replace the period $\mathcal{P}$ by the height of a *special cycle* $\mathcal{Z}_r$ on the Shimura variety $\text{Sh}(G_r \times G_{r+1})$:

$$\langle \mathcal{Z}_r, \mathcal{Z}_r \rangle_{\text{Nek}} \doteq L^{(r)}\left(\frac{1}{2}, \pi \times \pi'\right).$$

For $E/\mathbb{Q}$ of weight-2 newform f , the $\text{GL}(2)$ setting embeds into $U(r) \times U(r+1)$ via the base change $f_K = f \otimes \chi_K$. The special cycle $\mathcal{Z}_r$ corresponds to Heegner-type cycles in the r -fold Kuga-Sato variety, and the height pairing becomes the Nekovář height $\det(H_{ij})$.

Theorem 4 (Arithmetic GGP $\Rightarrow$ GGZ_r). *If the arithmetic GGP conjecture holds for $U(r) \times U(r+1)$, then GGZ_r holds for E .*

Proof. The arithmetic GGP gives $\langle \mathcal{Z}_r, \mathcal{Z}_r \rangle \doteq L^{(r)}(1/2, f_K \times f_K) = L^{(r)}(E/K, 1)$. Leibniz factorization ([1], Lemma 4) with $L(E_K, 1) \neq 0$ gives $L^{(r)}(E/K, 1) = L^{(r)}(E, 1) \cdot L(E_K, 1)$. The cycle $\mathcal{Z}_r$ decomposes under the Heegner hypothesis into the r derived classes $\kappa^{(i)}$; their height matrix equals $\det(H_{ij})$ by definition. Hence $\det(H_{ij}) \doteq L^{(r)}(E, 1)$, which is GGZ_r. $\square$

4 Known Cases

r	Shimura variety	Result	Reference
1	$U(1) \times U(2) \simeq \mathrm{GL}_2$ Shimura curve	Gross-Zagier	[2]
2	$U(2) \times U(3)$, diagonal in $E \times E \times E$ (under split-multiplicative hypothesis)	Darmon-Rotger	[3]
≥ 3	$U(r) \times U(r+1)$, r -fold Kuga-Sato	Open	—

Remark 5 (Why $r = 2$ required new ideas). Darmon-Rotger construct a diagonal class in $H^3(E \times E \times E)$ via the Gross-Kudla-Schoen cycle and use Hida theory (p -adic families of modular forms) to link it to the p -adic L -function $L_p(f \otimes f \otimes g)$ for an Eisenstein series g . The split-multiplicative hypothesis is used to normalize the p -adic period. For $r \geq 3$, the cycle lives in $H^{2r-1}(E^r)$; no p -adic L -function for $\mathrm{GL}(2)^r$ at $r \geq 3$ is known with the necessary interpolation properties.

5 The Open Step for $r \geq 3$

Conjecture 6 (Arithmetic GGP for $U(r) \times U(r+1)$). *Let $\mathcal{Z}_r$ be the r -th arithmetic diagonal cycle on $\mathrm{Sh}(U(r) \times U(r+1))$ defined via the r -fold Kuga-Sato variety. Then*

$$\langle \mathcal{Z}_r, \mathcal{Z}_r \rangle_{\mathrm{Nek}} = C_{r,E,K} \cdot \frac{L^{(r)}(E/K, 1)}{\Omega_{E/K}^r}$$

for an explicit nonzero constant $C_{r,E,K} \in \mathbb{Q}^\times$.

The proof of Conjecture 6 for general r requires:

1. **Construction of $\mathcal{Z}_r$.** The r -fold Kuga-Sato variety $\mathcal{W}_r \rightarrow \mathcal{M}$ (over the modular curve) carries a natural algebraic cycle in codimension r . For $r \geq 3$, the motivic cohomology class of $\mathcal{Z}_r$ is not yet controlled; the Beilinson conjectures predict it is non-trivial but do not construct it.
2. **p -adic L -function for $\mathrm{GL}(2)^r$.** An Euler system for $U(r) \times U(r+1)$ requires a p -adic L -function interpolating $L^{(r)}(E, s)$ near $s = 1$. For $r = 2$, this is the Garrett-Hida triple-product L -function. For $r \geq 3$, the analogue (Garrett L -function for r -tuples) is not constructed with the required interpolation properties.

3. **Explicit reciprocity law.** The height $\langle \mathcal{Z}_r, \mathcal{Z}_r \rangle_{\text{Nek}}$ must be linked to the value of the p -adic L -function via a Coleman map / Perrin-Riou logarithm. For $r = 1$, this is the Gross-Zagier formula itself. For $r = 2$, it is Darmon-Rotger's main theorem. For $r \geq 3$, this is open.

6 Main Theorem and Implication Chain

Theorem 7 (BSD, conditional on arithmetic GGP and finite III). *Assume Conjecture 6 for $U(r) \times U(r+1)$ and that $\text{III}(E/\mathbb{Q})$ is finite. Then the full BSD formula (1) holds for every elliptic curve $E/\mathbb{Q}$ with $r_{\text{an}}(E) = r$.*

Proof. Theorem 4 gives GGZ_r . The rank identity $r = r_{\text{an}}$ follows from [1] Theorem 1. The full formula follows from Corollary 2. $\square$

Remark 8 (Template summary). • *Regularize:* $L_\alpha(E, s) = L(E, s) \exp(-\alpha(s-1)^2)$; analytic rank preserved. $\checkmark$

- *Bound:* Kolyvagin Euler system bounds rank from above; GGZ_r bounds from below. Rank ≤ 1 unconditional; rank ≥ 2 conditional on arithmetic GGP.
- *Compact:* Heath-Brown average rank $\leq 3/2$; high-rank curves are non-generic. $\checkmark$ (statistical)
- *Regularity:* Full BSD formula (1) once regulator $= L^{(r)}(E, 1)/\Omega_E^r$ is established. Conditional.

The single hard step is arithmetic GGP for $U(r) \times U(r+1)$, $r \geq 3$: constructing the r -th diagonal Shimura cycle and proving the explicit reciprocity law.

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